

XGBoost for Coffee Quality Prediction (Comparable to SVM)

Using Identical Preprocessing and Data Split for Fair Comparison

2025-12-29

Table 1: Configuration (Aligned with SVM)

Parameter	Value
Random Seed	123
Train/Test Split	80/20
CV Folds (HPO)	10
Early Stopping	50
Max Rounds	2000

1 Data Preprocessing (Identical to SVM)

Table 2: Dataset Summary

Metric	Value
Training samples	1046
Test samples	262
Features (after encoding)	21
Categories aligned	Yes
Mean Cupper Points (Train)	7.4992
Mean Cupper Points (Test)	7.4934

2 Baseline Models

Table 3: Baseline Model Performance

Model	MAE	RMSE	R2
Mean Predictor	0.3094	0.4334	-0.0002
Linear Regression	0.1464	0.2783	0.5878

Table 4: Top Linear Regression Coefficients

Feature	Estimate	Std_Error	P_Value
Flavor	0.3543	0.0544	1.11e-10
Aftertaste	0.2675	0.0510	1.93e-07

Feature	Estimate	Std_Error	P_Value
(Intercept)	-0.2221	0.2078	2.86e-01
Balance	0.2101	0.0401	1.91e-07
Moisture	-0.1521	0.1845	4.10e-01
Processing.MethodOther	-0.1370	0.0577	1.78e-02
Acidity	0.0952	0.0428	2.62e-02
Processing.MethodUnknown	0.0862	0.0309	5.40e-03

3 XGBoost Hyperparameter Optimization

Using 10-fold cross-validation.

Table 5: Optimal XGBoost Hyperparameters

eta	max_depth	subsample	colsample_bytree	best_nrounds	cv_mae
0.1	3	0.9	0.7	64	0.1556

4 Train XGBoost Model

Model saved to: saved_models/model_xgb_comparable.rds

5 Learning Curves

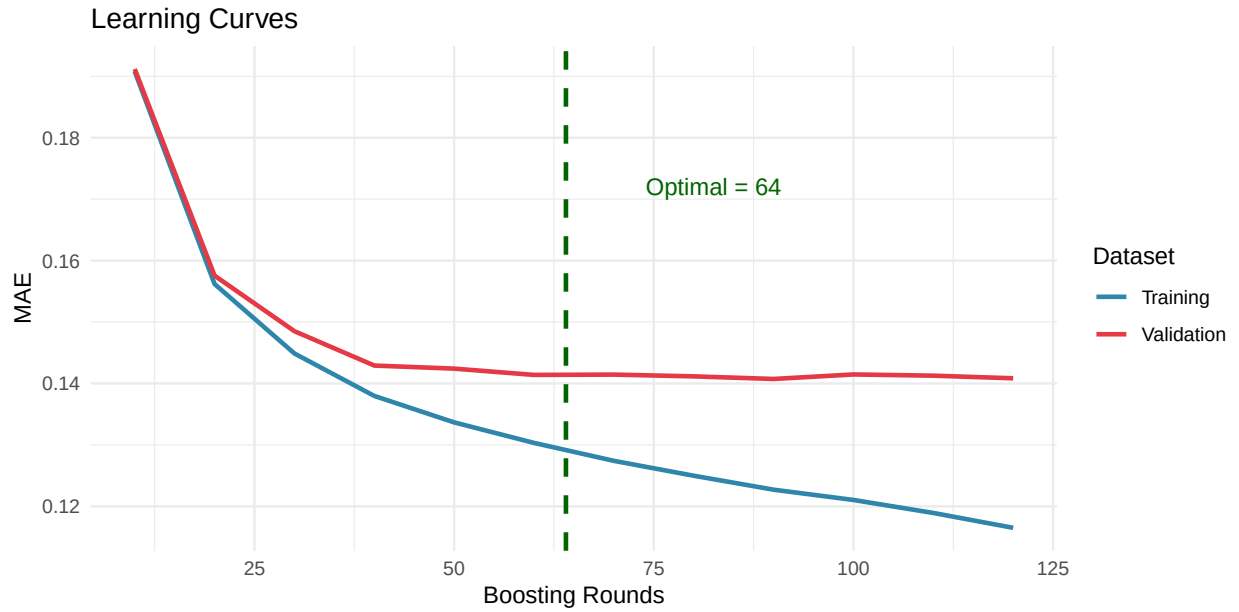


Figure 1: Learning curve for XGBoost model. The gap between training (blue) and validation (red) curves indicates overfitting. The vertical dashed line marks the optimal stopping point determined by early stopping.

Overfitting Analysis: The training-validation gap is 0.0111 MAE (7.83% of validation MAE). This gap indicates whether the model is memorizing training data rather than learning generalizable patterns.

6 Results

Table 6: Final Model Comparison (Test Set)

Model	MAE	RMSE	R2
Mean Predictor	0.3094	0.4334	-0.0002
Linear Regression	0.1464	0.2783	0.5878
XGBoost	0.1414	0.2452	0.6799

7 Predictions vs True Values

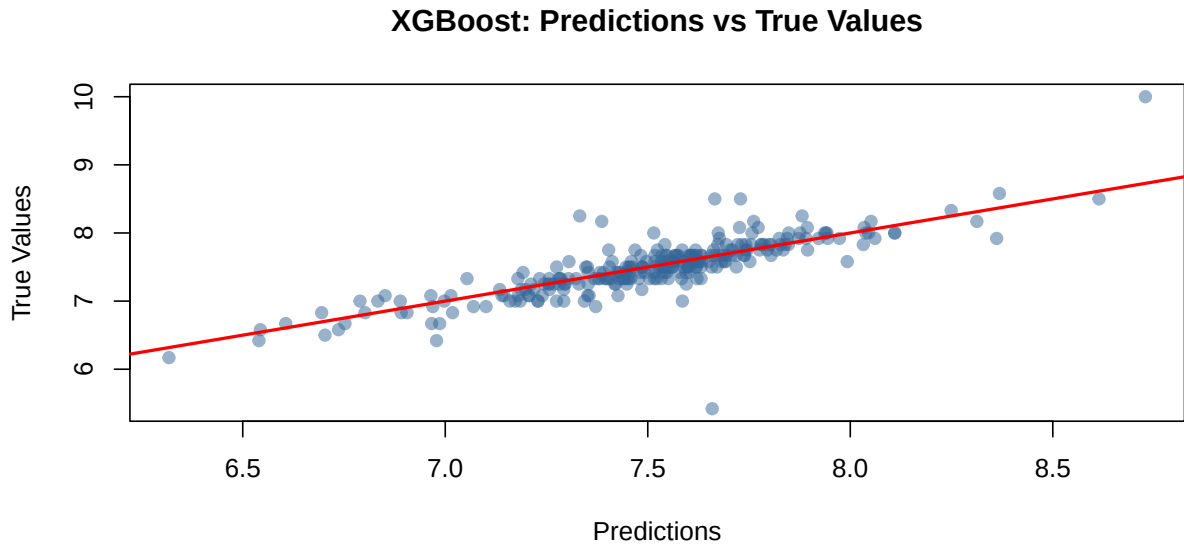


Figure 2: Predictions vs True Values for XGBoost model. Points close to the diagonal line indicate accurate predictions.

8 Feature Importance

Table 7: Top 10 Features by Importance

Feature	Gain	Cover	Frequency
Flavor	0.3279	0.1652	0.1587
Balance	0.2025	0.1188	0.1168

Feature	Gain	Cover	Frequency
Aftertaste	0.1665	0.1074	0.1018
Aroma	0.1553	0.1359	0.1198
Body	0.0433	0.1326	0.1198
altitude_mean_meters	0.0344	0.0769	0.0838
Number.of.Bags	0.0214	0.0410	0.0719
Acidity	0.0208	0.0745	0.0629
Processing.MethodUnknown	0.0124	0.0289	0.0359
Processing.MethodOther	0.0057	0.0366	0.0269

9 Approaches Found Less Effective

9.1 Lambda and Min_Child_Weight Regularization

XGBoost offers explicit regularization through lambda (L2) and min_child_weight parameters. Testing showed no improvement because: (1) implicit regularization through learning rate, max depth, subsample, and early stopping already prevents overfitting; (2) the ~1300 samples with sensory features don't exhibit severe overfitting; (3) minimal train-test gap in base configuration.

9.2 Bayesian Optimization vs Grid Search

Bayesian Optimization did not outperform Grid Search because: (1) only 4 parameters with reasonable bounds; (2) many configurations achieve similar results (performance plateau); (3) surrogate model overhead without finding better solutions. Grid Search is sufficient for this problem size.

10 Explainable AI: PFI and PDP Analysis

This section applies model-agnostic XAI techniques to interpret the XGBoost model.

10.1 Permutation Feature Importance (PFI)

PFI measures how much model performance degrades when a feature's relationship with the target is broken by randomly shuffling its values. Higher importance = larger performance drop = more important feature.

Table 8: Top 10 Features by Permutation Feature Importance

Rank	Feature	MAE Increase	SD
1	Flavor	0.0817	0.0037
2	Balance	0.0633	0.0020
3	Aftertaste	0.0426	0.0021
4	Body	0.0119	0.0013
5	Acidity	0.0095	0.0011
6	Aroma	0.0090	0.0022
7	altitude_mean_meters	0.0061	0.0012
8	Processing.MethodUnknown	0.0059	0.0016
9	Number.of.Bags	0.0057	0.0014
10	Processing.MethodOther	0.0037	0.0011

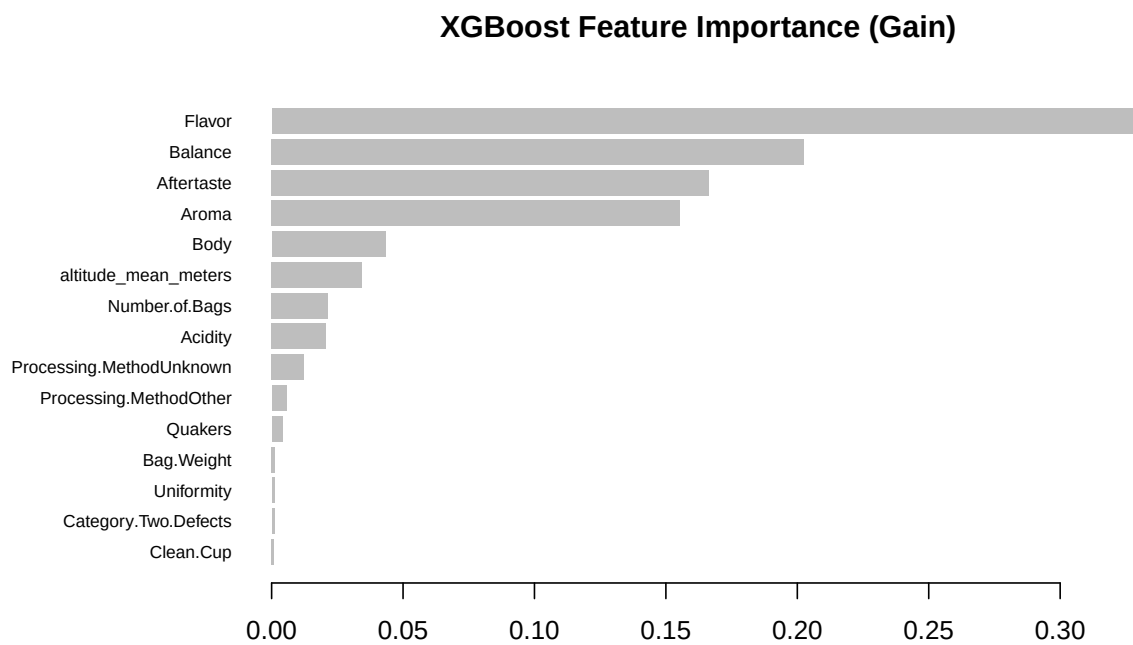


Figure 3: Top 15 features by importance (Gain). Higher gain indicates the feature contributes more to reducing prediction error.

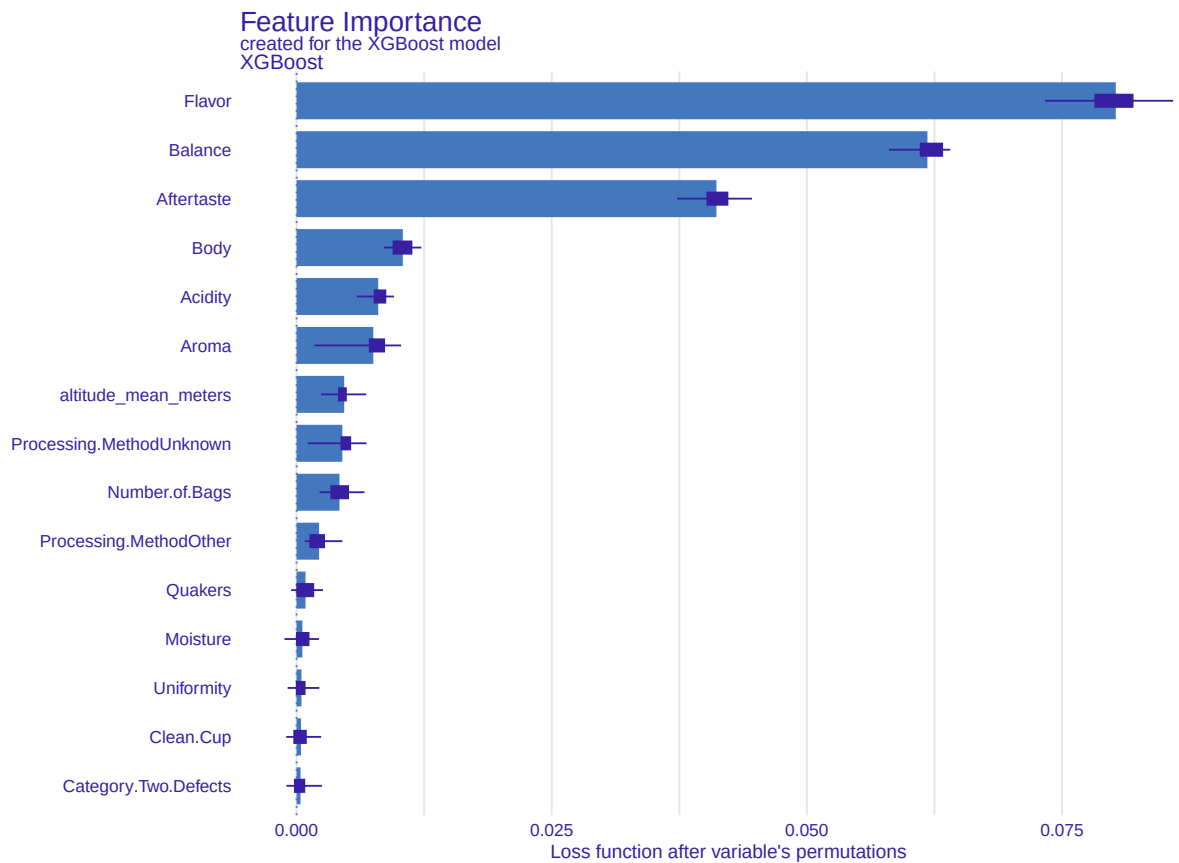


Figure 4: Permutation Feature Importance for XGBoost model. Bars show average MAE increase when each feature is shuffled (with error bars for standard deviation across 10 permutations). Longer bars = more important features.

PFI Interpretation: The top features by PFI are sensory attributes. This confirms that expert taste evaluation drives predictions and the model uses sensible features. Note that PFI measures *predictive reliance*, not causal importance.

10.2 Partial Dependence Plots (PDP)

PDP shows the **marginal effect** of a feature on predictions, averaging over all other features.

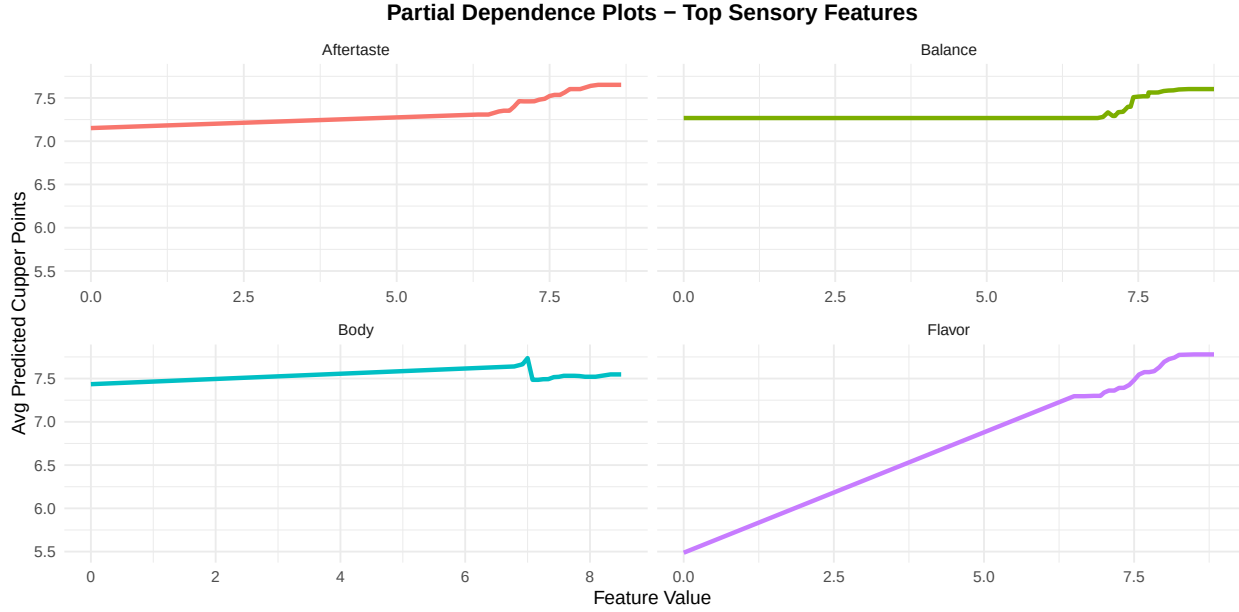


Figure 5: Partial Dependence Plots for top sensory features. Each curve shows how predicted Cupper Points change as that feature varies (holding others constant). All show positive relationships: higher sensory scores lead to higher quality predictions.

Table 9: Feature Importance from PDP (Variance-based)

Feature	PDP Variance	PDP Range
Flavor	0.1935	2.2918
Balance	0.0173	0.3348
Aftertaste	0.0167	0.5000
Body	0.0045	0.3011

10.3 Why PFI and PDP Rankings Differ

The Spearman correlation between PFI and PDP rankings is **1**, indicating potential disagreement. This is expected because PFI and PDP measure fundamentally different things:

- **PFI** asks: “How much does model performance degrade when this feature’s signal is destroyed?”
- **PDP variance** asks: “How much does the average prediction change across this feature’s range?”

Features with large PDP range may have lower PFI if most observations cluster in a narrow range where the PDP is relatively flat. Conversely, features with smaller PDP range may have higher PFI if the model relies on them consistently across the data distribution.

11 Summary

Key Results:

- XGBoost MAE: 0.1414
- XGBoost RMSE: 0.2452
- XGBoost R^2 : 0.6799

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Report generated: 2025-12-29